Research Topic Proposal

Markovian Embedding: A Way To Bypass The Memory Effect

A study on macro-financial data

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Abstract

Many macroeconomic and financial time series exhibit long memory, path dependence, or state-dependent dynamics that violate the Markov property, which severely limits the applicability of the powerful theoretical and computational toolbox built around Markov processes (filtering, dynamic programming, stationary-distribution analysis, efficient simulation). *Markovian embedding* is the general strategy of enlarging the state space of a non-Markovian process by appending auxiliary variables so that the resulting augmented process recovers the Markov property. This proposal has three objectives: (i) to present, in a mathematically rigorous manner, the theory of Markovian embeddings, ~~including finite-order embeddings, delay/Takens-type embeddings,~~ ~~and infinite-dimensional embeddings for long-memory processes~~; (ii) to motivate and justify their relevance for macro-financial time series, in which memory effects (volatility clustering, macro persistence, fractional integration) are pervasive; and (iii) to implement, benchmark, and document several competing embedding methodologies on the FRED-MD macroeconomic database, releasing the resulting code as an open, reproducible repository.

Keywords: Markovian Embedding, Representation Learning, Long Memory, State-Space Models, Time Series, FRED-MD

1. Problem Formulation

1.1. Setting and the memory effect

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $(X_t)_{t\in\mathbb{T}}$, $\mathbb{T} \in \{\mathbb{Z}, \mathbb{N}, \mathbb{R}_+\}$, be a stochastic process taking values in a measurable state space $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$, typically $\mathcal{X} \subseteq \mathbb{R}^d$. Denote by $\mathcal{F}_t^X = \sigma(X_s : s \leq t)$ the natural filtration generated by (X_t).

Definition 1.1 (Markov process)**.** *(X_t) is* Markovian *with respect to $(\mathcal{F}_t^X)$ if for every t and every Borel set $A \in \mathcal{B}(\mathcal{X})$,*

$$\mathbb{P}\left(X_{t+1} \in A \mid \mathcal{F}_t^X\right) = \mathbb{P}\left(X_{t+1} \in A \mid X_t\right) \quad \mathbb{P}\text{-}a.s.$$

Most macro-financial series fail this property: conditional on the present value alone, the future distribution still depends on the past trajectory. We refer to this as the *memory effect*. Two canonical sources of memory effect are of interest here:

- **Finite-order memory.** $X_{t+1} \mid \mathcal{F}_t^X \overset{d}{=} X_{t+1} \mid (X_t, X_{t-1}, \dots, X_{t-p+1})$ for some finite $p \geq 1$ (e.g. an AR(p) or GARCH(p, q) process).
- **Long memory.** The autocovariance function $\gamma(h) = \mathrm{Cov}(X_t, X_{t+h})$ decays hyperbolically, $\gamma(h) \sim c\, h^{2d-1}$ as $h \to \infty$ for some $d \in (0, 1/2)$, so that $\sum_h |\gamma(h)| = \infty$ (e.g. fractionally integrated processes, ARFIMA, fractional Brownian motion). No finite truncation of the past is a sufficient statistic for the future.

1.2. Markovian embeddings

Definition 1.2 (Markovian embedding). *Let $(\mathcal{S}, \mathcal{B}(\mathcal{S}))$ be a measurable* state space *(the* embedding space*). A* Markovian embedding *of (X_t) is a pair $(f, (S_t))$ where*

$$f : \mathcal{X}^{\mathbb{T}_{\leq t}} \longrightarrow \mathcal{S}, \qquad S_t := f\big((X_s)_{s \leq t}\big),$$

is a (measurable, causal) map from the path space up to time t into $\mathcal{S}$, such that:

(i) ***Markov property:*** *(S_t) is Markovian with respect to its own natural filtration $\mathcal{F}_t^S = \sigma(S_s : s \leq t)$, i.e.*

$$\mathbb{P}\left(S_{t+1} \in B \mid \mathcal{F}_t^S\right) = \mathbb{P}\left(S_{t+1} \in B \mid S_t\right), \qquad \forall B \in \mathcal{B}(\mathcal{S});$$

(ii) ***Consistency (recoverability):*** *there exists a measurable readout map $g : \mathcal{S} \to \mathcal{X}$ such that $X_t = g(S_t)$ a.s. for all t, i.e. f does not discard information about X itself, only reorganizes it.*

Remark 1.1 (The trivial embedding and the need for minimality). *Setting $S_t := (X_s)_{s \leq t}$ (the whole path, or equivalently $\mathcal{F}_t^X$ itself) trivially satisfies Definition 1.2, since $\mathbb{P}(S_{t+1} \mid \mathcal{F}_t^S) = \mathbb{P}(S_{t+1} \mid S_t)$ holds by construction ($\mathcal{F}_t^S = \mathcal{F}_t^X$). This embedding is useless in practice: it does not reduce dimensionality nor yield a tractable state. The object of interest is therefore not embeddability* per se *(which is always trivially true) but the existence of a* low-dimensional*, or at least* finite-dimensional*, embedding.*

Definition 1.3 (Order and minimality). *The* order *of an embedding is $\dim(\mathcal{S})$ when $\mathcal{S} \subseteq \mathbb{R}^k$ (or, more generally, a notion of statistical complexity of $\mathcal{S}$, e.g. its entropy rate or covering number). A Markovian embedding $(f^\star, S_t^\star)$ is* minimal *if for every other valid embedding (f, S_t) there exists a measurable h with $S_t^\star = h(S_t)$ a.s.; equivalently, $S_t^\star$ is a* minimal sufficient statistic *of $\mathcal{F}_t^X$ for predicting X_{t+1}.*

1.3. Canonical constructions

Example 1.1 (Finite-order embedding). *If (X_t) has finite-order memory p (Section 2.1), the map*

$$S_t := (X_t, X_{t-1}, \dots, X_{t-p+1}) \in \mathcal{X}^p$$

defines a finite-dimensional Markovian embedding: this is the classical "stacking" trick reducing an AR(p) to a first-order vector autoregression VAR(1) on $\mathcal{X}^p$.

Example 1.2 (Delay embedding / Takens' theorem)**.** *For a deterministic dynamical system observed through a scalar nonlinear map, Takens' embedding theorem guarantees that, generically, the delay vector* $S_t = (X_t, X_{t-\tau}, \ldots, X_{t-(m-1)\tau})$ *is a diffeomorphic (hence Markovian, in fact deterministic) embedding of the underlying* m*-dimensional attractor as soon as* $m \geq 2\,d_A + 1$*, where* d_A *is the box-counting dimension of the attractor (Takens, 1981). This motivates data-driven, delay-based embeddings even absent a known finite AR order.*

Example 1.3 (Latent/hidden-state embedding)**.** (X_t) *may be a noisy functional of an unobserved Markov chain* (Z_t) *(a hidden Markov model or linear-Gaussian state-space model):* $Z_{t+1} \mid \mathcal{F}_t^Z \overset{d}{=} Z_{t+1} \mid Z_t$ *and* $X_t = g(Z_t, \varepsilon_t)$*. Then* $S_t := Z_t$ *(or its filtering distribution / belief state* $\pi_t := \mathbb{P}(Z_t \in \cdot \mid \mathcal{F}_t^X)$*, by the Markov property of the filter, Durbin and Koopman, 2012) is a Markovian embedding of* (X_t)*, generally of much lower order than the raw history.*

Example 1.4 (Infinite-dimensional embedding of long-memory processes)**.** *When* $\sum_h |\gamma(h)| = \infty$*, no* finite-*order embedding of the form above exists, since finite differencing cannot annihilate hyperbolic autocovariance decay. However, long-memory Gaussian processes such as fractional Brownian motion* B^H *(*$H \in (0,1)$*) admit* Markovian representations *as superpositions (integral mixtures) of infinitely many Ornstein–Uhlenbeck (OU) processes with a continuum of mean-reversion speeds:*

$$X_t = \int_0^\infty Y_t(\theta)\,\mu(d\theta), \qquad dY_t(\theta) = -\theta\,Y_t(\theta)\,dt + dW_t,$$

for a suitable spectral measure μ *(Carmona and Coutin, 1998; Muravlev, 2011). The joint process* $(Y_t(\theta))_{\theta>0}$ *is Markovian in an infinite-dimensional (Hilbert-space-valued) state, and* finite-rank truncations *(a finite grid of* θ_i*, or Markovian semigroup approximations as in rough volatility models, Abi Jaber et al., 2019) provide practically tractable, approximately Markovian embeddings with a quantifiable approximation error. This is the central theoretical object motivating the use of Markovian embeddings for macro-financial series with long memory (e.g. realized volatility, inflation persistence).*

Proposition 1.1 (Existence vs. tractability trade-off)**.** *For any process* (X_t) *and any* $\varepsilon > 0$*, there exists a Markovian embedding* $(f_\varepsilon, S_t^\varepsilon)$ *with* $\dim(\mathcal{S}_\varepsilon) < \infty$ *such that*

$$\sup_t \, d_{TV}\Big(\mathbb{P}(X_{t+1} \in \cdot \mid \mathcal{F}_t^X),\ \mathbb{P}(X_{t+1} \in \cdot \mid S_t^\varepsilon)\Big) \leq \varepsilon,$$

whenever the memory of (X_t) *is* summable *in an appropriate mixing sense (e.g.* β*-mixing with summable coefficients). No such uniform finite-dimensional approximation exists, in general, without a rate assumption on the memory decay.*

Proposition 1.1 (to be proven rigorously in the literature-review phase, building on mixing-coefficient theory, Doukhan, 1994) frames the practical question addressed by this proposal: *which finite-dimensional embeddings achieve the best bias (approximation error, per Proposition 1.1) / variance (estimation error, given finite macro-financial samples) trade-off on real data?*

2. Methodology Suggested

The project is organized into five sequential work packages (WP).

WP1. Literature review. Identify and study the necessary background: Markov process theory and filtration/sufficiency theory; state-space models and Kalman/particle filtering; hidden Markov models; delay embeddings (Takens' theorem) and dynamical-systems reconstruction; long-memory and fractional processes (ARFIMA, fractional Brownian motion) and their Markovian (semimartingale/superposition) representations; rough volatility models; and representation learning approaches to state construction (recurrent neural networks, state-space deep learning models such as S4/Mamba, reservoir computing/echo-state networks), which can be read as *learned* Markovian embeddings.

WP2. Synthesis. Write a self-contained theoretical synthesis of the literature review, formalizing the definitions of Section 1, proving Proposition 1.1 (or a suitable variant) rigorously, and classifying existing embedding methods along two axes: (a) model-based vs. data-driven/learned, and (b) finite-order vs. infinite-dimensional/approximate.

WP3. Model selection. From the synthesis, select and justify up to four candidate Markovian embedding methodologies applicable to (multivariate) macro-financial time series, e.g.:

- finite-order VAR/vector-embedding (baseline);
- linear-Gaussian state-space / Kalman-filter embedding (dynamic factor model);
- Markovian (multi-factor OU) approximation of long-memory/rough-volatility dynamics;
- a learned neural embedding (recurrent or state-space neural network) trained to satisfy an approximate Markov / next-state predictive-sufficiency criterion.

Selection criteria: theoretical justification (Section 1), interpretability, computational tractability, and suitability to the sample size and sampling frequency of FRED-MD.

WP4. Implementation. Implement the selected models in a public, version-controlled Python repository (hosted on GitHub), with a common interface (fit / embed / predict), unit tests, and reproducible experiment scripts.

WP5. Reporting. Write a full paper presenting the theory, the model choices, the implementation, and the empirical benchmark, following standard academic structure (introduction, related work, theory, methodology, results, discussion, conclusion), with a complete bibliography.

3. Expected Outputs

1. A detailed, mathematically rigorous synthesis of Markovian embedding theory and its supporting literature (Section 1, extended and referenced).

2. A clear, justified overview of potential applications of Markovian embeddings to macro-financial data, covering both macroeconomic persistence (FRED-MD-type series) and financial memory effects (volatility clustering, rough volatility).

3. A clean, documented, open-source GitHub repository implementing a Python experimental pipeline that benchmarks (up to four) Markovian embedding methods on the FRED-MD

macroeconomic dataset (McCracken and Ng, 2016), including data preprocessing (stationarity transforms as in FRED-MD's transformation codes), embedding construction, and quantitative benchmark metrics (e.g. out-of-sample predictive log-likelihood, forecast RMSE, and empirical tests of the Markov property such as conditional independence / mixing diagnostics).

4. A full written paper consolidating the above, with all claims, definitions, and stated results properly referenced via a BibTeX bibliography (`references.bib`).

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